Project Name: Limitations and Future Scope

1. Current Limitations

While the proposed system provides a structured approach to delivering weather-based farming advice, the current version has several limitations:

- Limited Data Sources:
  The system may rely on basic or publicly available weather APIs, which may not always provide highly accurate or hyper-local village-level data.
- Prototype-Level Database:
  For initial implementation, data storage may use simple databases (e.g., SQLite or local storage). A production system would require scalable cloud databases.
- Basic Farmer Registration:
  Authentication mechanisms may be simple and not integrated with government databases like Aadhaar or agricultural records.
- Manual Officer Updates:
  Agricultural officers may need to manually input recommendations, which can delay real-time updates.

2. Technical Limitations

- **Internet Dependency:**
  The system requires internet connectivity, which may be unreliable in rural villages of Rajasthan.
- **Scalability Issues:**
  The prototype may not handle a large number of users across multiple districts efficiently without backend optimization.
- **Limited Personalization:**
  Recommendations may be generalized and not fully customized for each farmer's soil type, crop variety, or land size.
- **No Real-Time Sensor Integration:**
  The system does not currently integrate IoT-based weather sensors for real-time field-level data.

3. Future Enhancements (Roadmap)

A. AI-Based Crop Recommendation

- Use Machine Learning models to analyze:
  - Weather patterns
  - Soil conditions
  - Crop history
- Provide highly accurate and personalized crop suggestions.

B. Mobile Application Development

- Develop Android/iOS apps for farmers:
  - 📱 Voice-based alerts in local languages
  - 📍 GPS-based village-level weather updates
  - 🔔 Push notifications for extreme weather

C. IoT Integration

- Connect with smart farming sensors:
  - Soil moisture sensors
  - Temperature and humidity sensors
- Enable real-time monitoring and automated alerts.

D. Multi-Language Support

- Support regional languages like:
  - Hindi
  - Rajasthani
- Improve accessibility for rural farmers.

E. Government & Satellite Data Integration

- Integrate with:
  - IMD (Indian Meteorological Department)
  - Satellite weather data
- Improve prediction accuracy.

4. Real-World Deployment Considerations

- **Data Privacy & Security:**
  Farmer data must be securely stored and protected.
- **User Adoption Challenges:**
  Farmers may prefer traditional methods; awareness programs will be needed.
- **Training & Support:**
  Provide training sessions for farmers to effectively use the system.
- **Cost Constraints:**
  Implementation should remain low-cost to ensure accessibility.